Kitsune-Based Anomaly Detection: Technical Analysis

Arnav Inamdar

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1 Introduction

This document provides a technical overview of how the Kitsune-based model detects anomalies within the Unified DoS/DDoS Detection System and the feature that are used in the model. It explains the step-by-step detection process and provides a detailed breakdown of the features utilized.

2 Detection Process: Step-by-Step

Kitsune operates as an unsupervised anomaly detection system. It learns what "normal" traffic looks like and flags traffic that deviates significantly from that learned baseline.

2.1 Step 1: Traffic Interception & Feature Extraction

- **Input:** Raw network traffic (PCAP/PCAPNG files) or live packet capture.
- **Process:**
  1. Raw packets are read.
  2. Packets are aggregated into **bidirectional flows** (sessions between two IP addresses and ports).
- **Output:** A numerical feature vector x representing the flow behavior.

2.2 Step 2: Model Inference (Reconstruction)

- **Input:** Feature vector x.
- **Process:**
  1. The vector x is passed into the **Autoencoder** neural network.
  2. The Autoencoder compresses the input into a lower-dimensional representation (encoding).
  3. The network reconstructs the original input from this compressed representation.
- **Output:** Reconstructed vector x'.

2.3 Step 3: Anomaly Scoring (RMSE)

- **Logic:**
  - If x represents **benign traffic**, the model (trained on benign data) reconstructs it accurately. The difference between x and x' is small.
  - If x represents **anomalous traffic**, reconstruction accuracy decreases because such patterns were not observed during training. The difference between x and x' becomes large.
- **Calculation:** The **Root Mean Square Error (RMSE)** is computed between the input and reconstruction:

$$\text{Score} = \text{RMSE}(x, x') \tag{1}$$

2.4 Step 4: Thresholding & Verdict

- **Input:** Anomaly Score (RMSE).
- **Process:**
  1. The score is compared against a predefined threshold τ.
- **Verdict:**
  - **If Score $> \tau$:** Traffic is flagged as an **Anomaly**.
  - **If Score $\leq \tau$:** Traffic is considered **Benign**.

3 Feature Analysis

The model relies on **33 statistical features** extracted from each flow. These features capture traffic volume, timing characteristics, and packet behavior.

3.1 Packet Counts (Volume)

- **Flow Duration:** Total time the connection was active (microseconds).
- **Total Fwd Packets:** Number of packets sent by the initiator.
- **Total Backward Packets:** Number of packets sent by the responder.
- **Flow Bytes/s:** Data transfer rate (bytes per second).
- **Flow Packets/s:** Packet transfer rate (packets per second).

Significance: High packet rates or counts within short durations are classical indicators of flooding attacks (UDP Flood, ICMP Flood).

3.2 Packet Lengths (Size Dynamics)

These features measure payload size distribution.

- **Total Length of Fwd/Bwd Packets**
- **Fwd/Bwd Packet Length Max**
- **Fwd/Bwd Packet Length Min**

- **Fwd/Bwd Packet Length Mean**
- **Fwd/Bwd Packet Length Std**

Significance:

- **DDoS:** Often uses fixed-size packets (near-zero standard deviation).
- **Normal Traffic:** Typically exhibits variable packet sizes (higher standard deviation).
- **Data Exfiltration:** May show unusually large packets (high max/mean).

3.3 Inter-Arrival Times (IAT) (Timing)

These features measure the time gap between consecutive packets.

- **Flow IAT Mean/Std/Max/Min**
- **Fwd IAT Total/Mean/Std/Max/Min**
- **Bwd IAT Total/Mean/Std/Max/Min**

Significance:

- **Automated Attacks:** Often show uniform timing (low standard deviation) or extremely short gaps (low mean).
- **Human/Normal Traffic:** Typically bursty and irregular (higher standard deviation).

3.4 TCP Flags (Behavior)

Counts of specific TCP control flags set within packets.

- **Fwd/Bwd PSH Flags**
- **Fwd/Bwd URG Flags**

Significance: Abnormal flag usage may indicate scanning, OS fingerprinting, or specific TCP flood patterns (e.g., PSH+ACK floods).

4 Proposed Features for Fine-Tuning

To improve the model's accuracy against State-Exhaustion and Slow-Rate attacks the following features were some of the optional ones which can be added later on after primary approvals.

4.1 A. Critical TCP Flags (SYN, ACK, RST, FIN)

Features:

- SYN Flag Cnt
- ACK Flag Cnt
- RST Flag Cnt
- FIN Flag Cnt

Reason for Addition:

- **SYN Floods:** Explicit SYN counting enables detection of SYN-heavy, ACK-light traffic patterns typical of half-open connection attacks.
- **ACK Floods:** High ACK counts without corresponding data may indicate reflection or state-table exhaustion attacks.
- **Scanning Activity:** Elevated RST or FIN counts strongly indicate port scanning or reconnaissance phases preceding DDoS attacks.

4.2 B. TCP Window Size

Features:

- Init Fwd Win Byts
- Init Bwd Win Byts

Reason for Addition:

- **Bot Fingerprinting:** TCP window size is determined by the operating system's TCP stack. Legitimate systems exhibit standardized patterns.
- **Detection Logic:** IoT botnets (e.g., Mirai) or scripted tools (e.g., hping3) often use hard-coded or non-standard window sizes, enabling differentiation between bots and legitimate clients.

4.3 C. Active / Idle Times

Features:

- Idle Mean
- Idle Std

Reason for Addition:

- **Slow-Rate Attacks (Slowloris):** These attacks maintain open connections by periodically sending partial requests.
- **Detection Logic:** While Flow Duration measures total session time, Idle Mean specifically exposes intentional waiting behavior typical of slow-rate attacks.